

Structure and Function of an Atom

Atom

- Atom is the basic unit of matter

It consists of 3 main subatomic particles

Protons

Neutrons

Electrons

Nucleus

- The Nucleus of an atom contains Protons and Neutrons

- Protons = Positively charged +

The number determines the element

- Neutrons = Negative/NO charge –

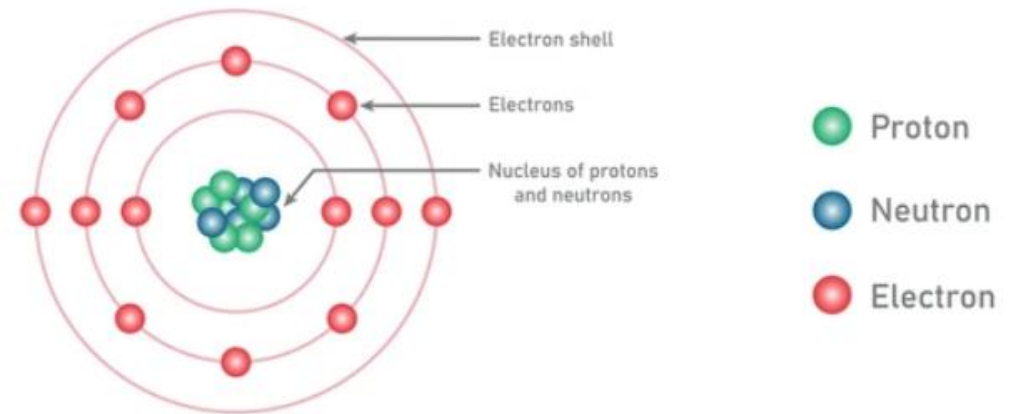
They contribute to the atom's mass

Electrons

Negatively charged and orbit the Nucleus in different **energy** levels/shells

Smaller than Protons and Neutrons

Determines the chemical properties of the element and how it interacts with other elements



Molecule

- A molecule is the smallest part of a compound
- It is made up of one or more atoms
- These atoms can be the same or different



Compounds

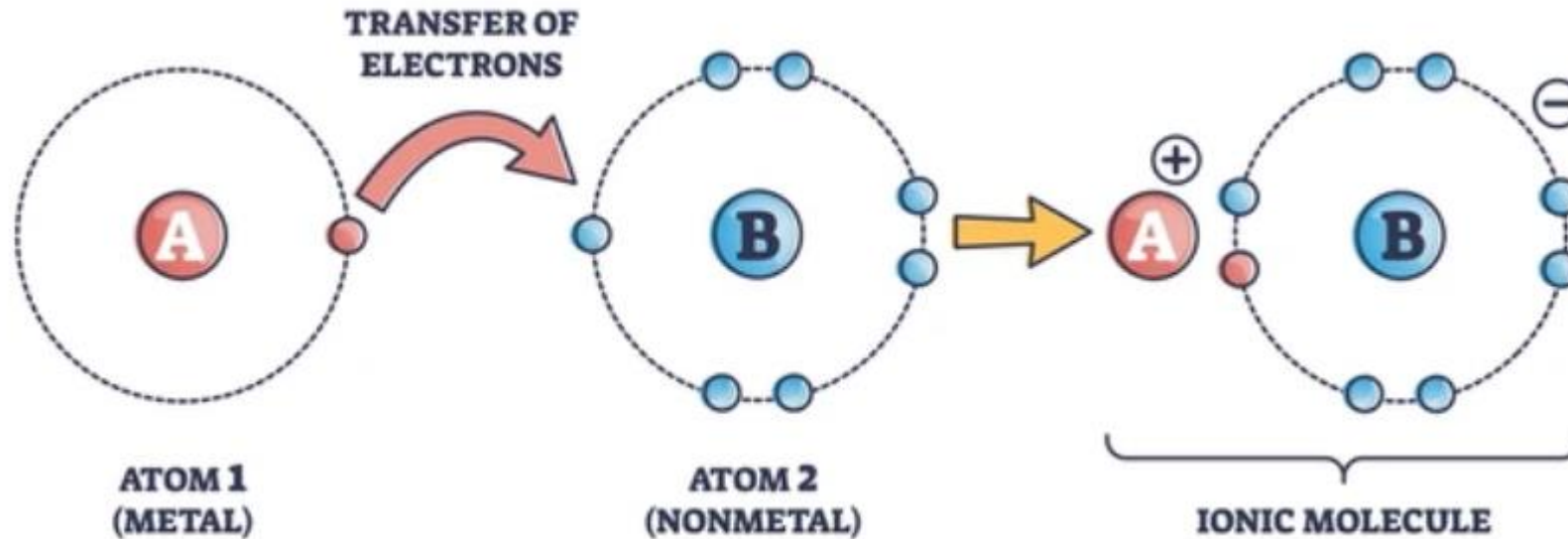
- A compound is a substance made up of two or more different elements that are chemically bonded together.
- These elements can be combined in specific ratios, and the bonding can involve sharing or transferring electrons.
- Compounds can be classified into two main categories based on the type of bond they have:
 - 1. Ionic Compounds:** These are formed when atoms transfer electrons from one to another. The result is the creation of positively and negatively charged ions that are attracted to each other.
 - 2. Covalent Compounds:** These form when two atoms share electrons. The shared electrons allow both atoms to achieve a more stable electron configuration. Examples of covalent compounds include **water (H₂O)** and **carbon dioxide (CO₂)**.

Ion

- An ion is an atom or molecule which has gained or lost one or more electrons
- They do this in order to achieve a more stable configuration
- When an atom loses an electron/s it becomes a **cation (positively charged)**
- When an atom gains an electron/s it becomes an **anion (negatively charged)**

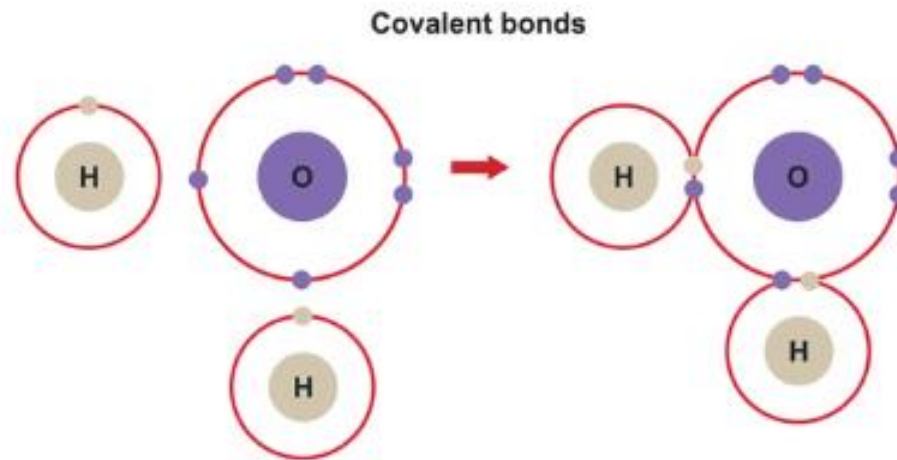
Ionic Bonding

- Type of chemical bonding formed from electrostatic attraction between oppositely charged ions



Covalent Bond

- Interatomic bonding that results from the sharing of a pair of electrons between 2 atoms
- The bond arises from the electrostatic attraction of the nuclei for the same electron



Key Concepts

- **Chemical Bonds:** The way atoms are held together in compounds. They can be ionic (based on opposite charges attracting) or covalent (based on shared electrons).
- **Molecular Formula:** This tells you how many atoms of each element are in a molecule of a compound. For instance, water has the formula H_2O , meaning 2 hydrogen atoms and 1 oxygen atom.
- **Chemical Reactions:** Compounds can be created or broken apart through chemical reactions, where bonds are made or broken.